

NT219 - Cryptography

Week 07-08: Modern Asymmetric Cryptography (P2)

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October 27, 2025

Outline

- 1 RSA Review
 - Raw encryption risks
 - OAEP Padding and reduce computation overhead using Carmichael function
 - Revise RSA-KEM (Encaps/Decaps)
- 2 Key Exchange Protocol (FFDHE)
- 3 Elliptic Curve Cryptography (ECC)
 - ECC group law
 - Key exchange protocol (ECDHE)
 - ECC Ciphers
 - Security Equivalences and why ECC?
- 4 PQS ML-KEM

Warm-up: RSA-based Agreeing an AES session key

- **RSA key transport:** the server uses its private key d to decrypt a client-chosen secret. No PFS unless using ephemeral RSA (rare).
- **Hack RSA raw for KEM:** $K = 16$ bytes, $K = 24$ bytes
- **RSA-OAEP Padding:**

$$EM = \text{EME-OAEP-Encode}(K, \text{label}), \quad c = EM^e \bmod n,$$

$$EM = 0x00 \parallel \text{maskedSeed} \parallel \text{maskedDB}$$

Thuật ngữ: EME-OAEP = Encoding Method for Encryption – Optimal Asymmetric Encryption Padding

RSA Key Generation (with CRT)

Goal. Create (n, e) and (d) with CRT parameters for fast decryption/signing.

Inputs: security parameter k (modulus size in bits, e.g., 2048 or 3072).

Algorithm

- 1 Sample distinct random primes p, q with $|p| \approx |q| \approx k/2$.
- 2 Compute modulus $n = pq$.
- 3 Compute Carmichael value $\lambda(n) = \text{lcm}(p-1, q-1)$.
- 4 Choose public exponent e with $\text{gcd}(e, \lambda(n)) = 1$ (commonly $e = 65537$).
- 5 Compute private exponent $d \equiv e^{-1} \pmod{\lambda(n)}$.
- 6 Precompute CRT params: $d_p = d \bmod (p-1)$, $d_q = d \bmod (q-1)$, $q_{\text{inv}} \equiv q^{-1} \pmod{p}$.
- 7 Output: public key (n, e) ; private key (d) plus $(p, q, d_p, d_q, q_{\text{inv}})$.

Notes. Use strong prime generation, reject close primes, and run consistency checks (e.g., test decryption of a known value).

Thuật ngữ: modulus n — mô-đun n ; Carmichael $\lambda(n)$ — hàm Carmichael $\lambda(n)$; public exponent — số mũ công khai; private exponent — số mũ bí mật; CRT parameters — tham số CRT; modular inverse — nghịch đảo mô-đun

RSA Encryption (Raw/Textbook)

Goal. Encrypt an integer message for the holder of (n, e) .

Inputs: public key (n, e) ; message integer m with $0 \leq m < n$.

Algorithm

- 1 Check range: ensure $0 \leq m < n$.
- 2 Compute ciphertext: $c \equiv m^e \pmod{n}$.
- 3 Output c .

Notes. This is *textbook* RSA (no padding); use only for demos/tests. In practice, apply a secure padding scheme (e.g., OAEP) before exponentiation.

Thuật ngữ: encryption — mã hoá; plaintext m — bản rõ m ; ciphertext c — bản mã c ; exponentiation — lũy thừa; modulus n — mô-đun n ; textbook (no padding) — bản thô (không đệm)

RSA Decryption (Raw, with CRT option)

Goal. Recover m from c using d (and optionally accelerate with CRT).

Inputs: private key d (or CRT params p, q, d_p, d_q, q_{inv}); ciphertext integer c with $0 \leq c < n$.

Algorithm (direct)

- 1 Check range: ensure $0 \leq c < n$.
- 2 Compute plaintext: $m \equiv c^{d_p} \pmod{n}$.
- 3 Output m .

Algorithm (CRT acceleration)

- 1 $m_p \equiv c^{d_p} \pmod{p}$, $m_q \equiv c^{d_q} \pmod{q}$.
- 2 $h \equiv (m_p - m_q) \cdot q_{inv} \pmod{p}$.
- 3 $m \equiv m_q + q \cdot h \pmod{n}$.
- 4 Output m .

Notes. Result $m \in [0, n-1]$. Optional check: verify $m^e \equiv c \pmod{n}$. This is raw RSA (no padding).

Thuật ngữ: decryption — giải mã; private exponent d — số mũ bí mật d ; CRT — định lý dư Trung Hoa; recombination — kết hợp; q_{inv} — nghịch đảo của q theo mô-đun p ; range check — kiểm tra miền giá trị

RSA-KEM (Encaps/Decaps) — RAW demo

Goal. Encapsulate a random secret under RSA, then derive symmetric keys via KDF (hybrid KEM–DEM style). **Encapsulate (recipient (n, e))**

- 1 Sample random r (bitlength $\approx$ security level).
- 2 $c \leftarrow r^e \bmod n$ (encapsulation).
- 3 $K \leftarrow \text{KDF}(\text{I2OSP}(r), \text{session info})$; output (c, K) .

Decapsulate (recipient d)

- 1 $r \leftarrow c^d \bmod n$.
- 2 $K \leftarrow \text{KDF}(\text{I2OSP}(r), \text{session info})$.

Security intuition. Insecure when $r^e < n$ (đặc biệt với e nhỏ).

Thuật ngữ: RSA-KEM — bao khoá bằng RSA; encapsulate/decapsulate — gói/giải gói; KDF — hàm dẫn xuất khoá; DEM/AEAD — lớp dữ liệu mã hoá (AEAD); one-wayness — tính một chiều, session info = Thông tin phiên

OAEP Encoding — Notation

Fixed parameters

- n : RSA modulus
- $k_N := \lceil \log_2 n \rceil / 8$ — **modulus length (bytes)**
- **Hash function** $H : \{0, 1\}^* \rightarrow \{0, 1\}^{hLen}$ (e.g., SHA-256) $\Rightarrow$ output length $hLen$ bytes
- **Mask Generation Function (MGF1)** based on H : $MGF1(Z, \ell) = \text{expand } H \text{ iteratively to produce } \ell \text{ bytes}$
- Optional label L (often empty)

Data layout

- $IHash = H(L)$ — hash of label L
- $DB = IHash \parallel PS \parallel 0x01 \parallel M$, where $|DB| = k_N - hLen - 1$
- **seed** $\in \{0, 1\}^{hLen}$ — uniformly random

Thuật ngữ: k_N — độ dài mô-đun (bytes); H — hàm băm (ví dụ SHA-256); MGF1 — hàm tạo mặt nạ dựa trên H ; $IHash = H(L)$ — băm nhãn; PS (padding string) — chuỗi toàn byte 0, độ dài sao cho DB đủ kích thước; seed/maskedSeed — hạt giống và hạt giống đã che; DB/maskedDB — khối dữ liệu trước và sau khi che.

RSA–OAEP Encryption (RFC 8017, Notation-Fixed)

Inputs: public (n, e) , message M with $mLen \leq k_N - 2 hLen - 2$, OAEP params $(H, \text{MGF1}, L)$.

Algorithm

- ① $IHash \leftarrow H(L)$.
- ② $DB \leftarrow IHash \parallel \text{PS} \parallel 0x01 \parallel M$ where $|DB| = k_N - hLen - 1$.
- ③ Sample uniform **seed** $\leftarrow \{0, 1\}^{hLen}$ bytes.
- ④ $\text{maskedDB} \leftarrow DB \oplus \text{MGF1}(\text{seed}, k_N - hLen - 1)$.
- ⑤ $\text{maskedSeed} \leftarrow \text{seed} \oplus H(\text{maskedDB})$.
- ⑥ $EM \leftarrow 0x00 \parallel \text{maskedSeed} \parallel \text{maskedDB}$ (exactly k_N bytes).
- ⑦ $c \leftarrow \text{OS2IP}(EM)^e \bmod n$; output $C \leftarrow \text{I2OSP}(c, k_N)$.

Remark. k_N is *byte length of n* , not a key and not the random seed.

Thuật ngữ: OS2IP/I2OSP — đổi chuỗi ↔ số nguyên; mLen bound — ràng buộc độ dài thông điệp; k_N is not a key — k_N không phải khoá; seed length = $hLen$ — seed dài $hLen$ byte

RSA Decryption with CRT + OAEP Decoding (Notation-Fixed)

Inputs: C (k_N bytes), private $(p, q, d_p, d_q, q_{\text{inv}})$, public (n, e) , OAEP params $(H, \text{MGF1}, L)$.

Algorithm

- ① $c \leftarrow \text{OS2IP}(C)$; optionally blind: $c' = c \cdot u^e \bmod n$.
- ② Exponentiate via CRT: $m_1 \leftarrow (c')^{d_p} \bmod p$, $m_2 \leftarrow (c')^{d_q} \bmod q$.
- ③ Recombine: $h \leftarrow (q_{\text{inv}}(m_1 - m_2)) \bmod p$; $m' \leftarrow m_2 + qh \pmod n$; unblind if used.
- ④ $\text{EM} \leftarrow \text{I2OSP}(m', k_N)$; parse $0x00 \parallel \text{maskedSeed} \parallel \text{maskedDB}$.
- ⑤ Recover seed = $\text{maskedSeed} \oplus H(\text{maskedDB})$.
- ⑥ $\text{DB} = \text{maskedDB} \oplus \text{MGF1}(\text{seed}, k_N - hLen - 1)$.
- ⑦ Verify $IHash$, find $0x01$ delimiter, output M ; else **DECODINGERROR**.

Constant-time checks for all comparisons and shape validations.

Thuật ngữ: CRT decryption — giải mã dùng CRT; unblinding — bỏ làm mù; delimiter $0x01$ — dấu tách $0x01$; decoding error — lỗi giải mã; constant-time checks — kiểm tra hằng thời gian

RSA-KEM (Encaps/Decaps) — Revision

Goal. Encapsulate a random secret under RSA, then derive symmetric keys via KDF (hybrid KEM-DEM).

Encapsulate (recipient (n, e))

- ➊ Set $k_N = \lceil \log_2 n \rceil / 8$. Sample $x \xleftarrow{\$} \{1, \dots, n-1\}$ *uniformly* (resample if $\gcd(x, n) \neq 1$).
- ➋ $c \leftarrow x^e \bmod n$ (encapsulation).
- ➌ $K \leftarrow \text{KDF}(\text{I2OSP}(x, k_N), \text{"RSA-KEM"})$; output (c, K) .

Decapsulate (recipient d)

- ➊ $x \leftarrow c^d \bmod n$.
- ➋ $K \leftarrow \text{KDF}(\text{I2OSP}(x, k_N), \text{"RSA-KEM"})$.

Security intuition. Reduces to RSA one-wayness over $\mathbb{Z}_n^*$ plus the KDF assumption. Use K with **DEM** = **AEAD** for data.

Thuật ngữ: uniform modulo n — chọn đồng đều theo mô-đun n ; small-exponent root — tấn công căn bậc e (tránh bằng cách chọn x theo n); k_N (modulus length) — độ dài mô-đun; AEAD (DEM) — lớp dữ liệu mã hoá

Attack Complexity — RSA vs. AES

Rule-of-thumb security equivalence

- RSA-1024 $\approx$ AES-80 (obsolete)
- RSA-2048 $\approx$ AES-112
- RSA-3072 $\approx$ AES-128
- RSA-7680 $\approx$ AES-192
- RSA-15360 $\approx$ AES-256

Complexity insight

- RSA (and DH) rely on **integer factoring** or **discrete logarithm** problems — solved fastest by the **Number Field Sieve (NFS)** with sub-exponential complexity.
- AES resists attack by design — best known generic attack is exhaustive **key search** with full exponential cost 2^k .
- Hence, asymmetric keys (RSA/DH) must be **much longer** to match a symmetric key's security.

Operational notes

- Use at least RSA-3072 for 128-bit equivalent security (NIST recommendation).

Hardness Assumptions for (Finite-Field) DHE

Here, “hard” means there is no known efficient (polynomial-time) algorithm to solve the problem for large parameters.

Let (p, g) with subgroup of order q .

- **Discrete Logarithm (DL):** given $y = g^x \bmod p$, find x .
- **Computational DH (CDH):** given (g^a, g^b) , compute g^{ab} .
- **Decisional DH (DDH):** given (g^a, g^b, g^c) , decide if $c \equiv ab \pmod{q}$.

Relations (easiness implications)

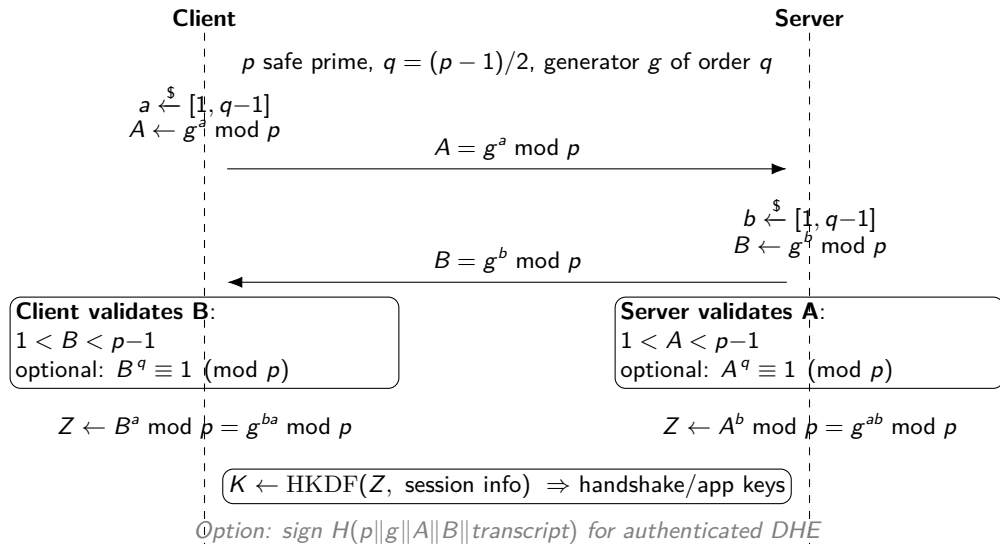
$DL \Rightarrow CDH \Rightarrow DDH$ (if you can solve DL, you can solve CDH, then DDH)

Security intuition

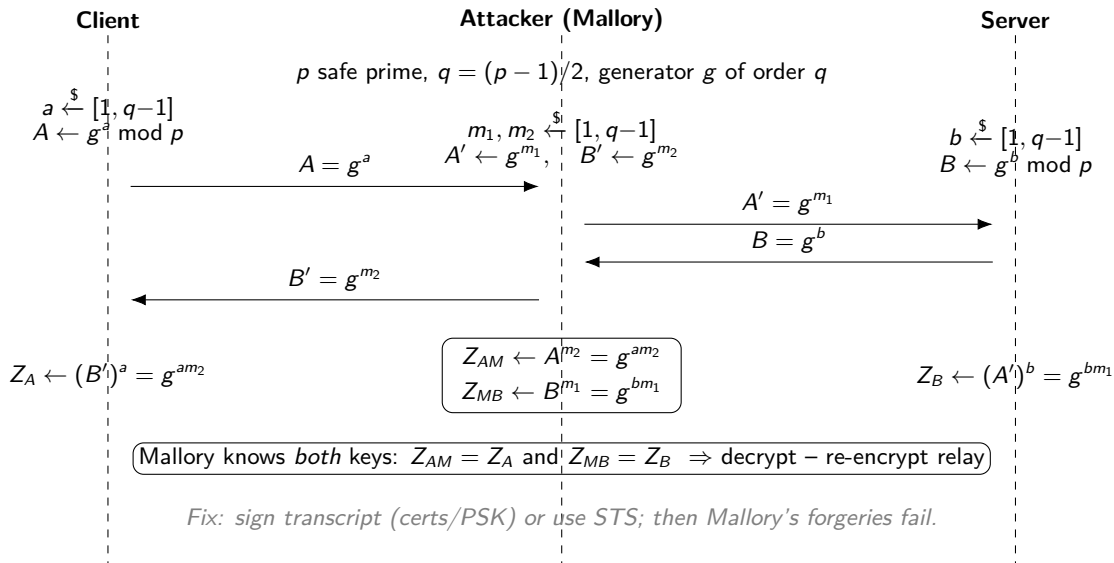
- In generic groups, DDH is **believed hard**.
- Using safe primes ($p=2q+1$) and generator of order q avoids parity leaks and small-subgroup traps.

Thuật ngữ: discrete logarithm (DL) — bài toán log rời rạc; computational DH (CDH) — DH tính toán; decisional DH (DDH) — DH quyết định; easiness implication — quan hệ dễ-khó; safe prime subgroup — nhóm con số nguyên tố an toàn

DHE Protocol (Ephemeral) — Client/Server Figure



DHE Man-in-the-Middle (No Authentication)



DHE vs RSA-KEM — Properties & Trade-offs

Property	DHE (ephemeral)	RSA-KEM
Main assumption	DL/CDH/DDH	factoring/RSA
Forward secrecy	Yes (ephemeral a, b)	No (unless ephemeral RSA keys)
Auth mechanism	signatures/PSK/certs	certs (RSA key) or signatures
Precomputation risk	fixed p aids NFS precomp	per-server modulus; no shared n
Round trips	1 extra for auth (varies)	similar (encaps only)
Performance	1 exp./side; server heavy	cheap encrypt (small e), costly decrypt
PQ resistance	broken by Shor	broken by Shor

When to choose

- Need PFS or better server compromise resilience: **DHE/ECDHE**.
- Legacy RSA PKI, simple KEM integration, no PFS requirement: **RSA-KEM**.

Thuật ngữ: forward secrecy — bí mật chuyển tiếp; precomputation — tiền tính; modulus — mô-đun; server-heavy decrypt — giải mã nặng phía máy chủ; PQ (post-quantum) — hậu lượng tử

Cheat Sheet — Key Sizes & Security Equivalences

AES Security	DHE modulus p	RSA modulus n	Typical Use Horizon
AES-112	2048	2048	short-term (to ~2030)
AES-128	3072	3072	mid-term (to ~2035–2040)
AES-192	7680	7680	long-term (beyond 2040)
AES-256	15360	15360	very long-term / top-secret

Prefer **ECDHE** for same-level security with smaller keys (e.g., P-256/X25519 ~ AES-128 ~ RSA-3072/FFDHE-3072).

Thuật ngữ: AES security level — mức an toàn AES; short/mid/long-term — ngắn/trung/dài hạn; NIST SP 800-57 Pt 1 Rev 5 — khuyến nghị chuẩn của NIST; equivalent security — mức an toàn tương đương; modulus size — kích thước mô-đun

From Algebra to Elliptic Curves

Algebraic ladder

- **Group** $(G, \oplus)$: closure, associativity, identity O , inverse
- **Ring** $(R, +, \cdot)$: two operations, distributivity
- **Field** $(\mathbb{F}, +, \cdot)$: every nonzero has a multiplicative inverse

Elliptic curve over field $\mathbb{F}$

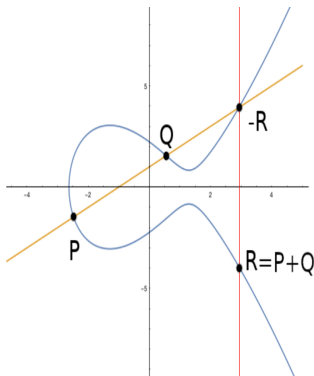
- Short Weierstrass: $E : y^2 = x^3 + ax + b$, $\Delta \neq 0$
- Identity O is the point at infinity; group law defines $P \oplus Q$ geometrically/algebraically.

Popular models

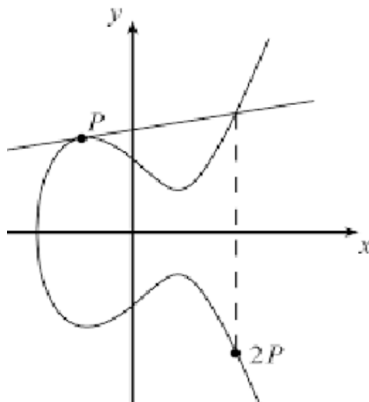
- **Weierstrass**: NIST P-256, P-384, ...
- **Montgomery**: Curve25519 (x-only ladder)
- **(Twisted) Edwards**: Ed25519 (fast, unified add)

Thuật ngữ: group — nhóm; ring — vành; field — trường; short Weierstrass model — mô hình Weierstrass ngắn; discriminant — biệt thức; point at infinity — điểm vô cực; point addition — phép cộng điểm; Montgomery ladder — thang Montgomery; (twisted) Edwards — Edwards xoắn

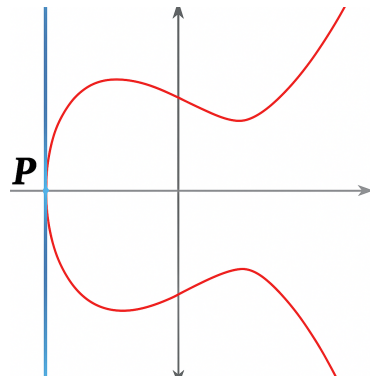
Elliptic group law



(a) ECC addition $P + Q$



(b) ECC point doubling $P + P$



(c) ECC point at infinity O ($P + P$ for $y_P = 0$)

Elliptic Curve Group Law — Weierstrass Form

Short Weierstrass model:

$$E : y^2 = x^3 + ax + b, \quad \Delta = -16(4a^3 + 27b^2) \neq 0$$

Group law (addition):

$$P = (x_1, y_1), \quad Q = (x_2, y_2), \quad R = P \oplus Q = (x_3, y_3)$$

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1}, & P \neq Q, \\ \frac{3x_1^2 + a}{2y_1}, & P = Q \end{cases}$$

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1$$

Identity element: O — the point at infinity. **Inverse:** $-(x, y) = (x, -y)$.

Montgomery Curves — Affine Group Law (minimal)

Model:

$$E : By^2 = x^3 + Ax^2 + x, \quad B(A^2 - 4) \neq 0$$

Affine group law:

$$P = (x_1, y_1), \quad Q = (x_2, y_2), \quad R = P \oplus Q = (x_3, y_3)$$

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1}, & P \neq Q, \\ \frac{3x_1^2 + 2Ax_1 + 1}{2By_1}, & P = Q \end{cases} \quad \begin{aligned} x_3 &= B\lambda^2 - A - x_1 - x_2, \\ y_3 &= \lambda(x_1 - x_3) - y_1 \end{aligned}$$

Neutral element & inverse:

$$O \text{ (point at infinity)}, \quad -(x, y) = (x, -y), \quad P \oplus (-P) = O.$$

Domain notes: làm việc trên trường đặc trưng $\neq 2$; cần $x_1 \neq x_2$ khi $P \neq Q$ và $y_1 \neq 0$ khi $P = Q$. X25519 dùng *x-only Montgomery ladder* trên tọa độ X/Z .

Thuật ngữ: neutral element O — phần tử đơn vị O ; inverse $-P$ — phần tử nghịch đảo $-P$; $-(x, y) = (x, -y)$ — đổi dấu y ; point at infinity — điểm vô cực; affine group law — luật nhóm affine

Elliptic Curve Group Law — Twisted Edwards Form

Twisted Edwards model:

$$E : ax^2 + y^2 = 1 + dx^2y^2, \quad a, d \neq 0$$

Group law (addition):

$$P = (x_1, y_1), \quad Q = (x_2, y_2), \quad R = P \oplus Q = (x_3, y_3)$$

$$x_3 = \frac{x_1y_2 + y_1x_2}{1 + dx_1x_2y_1y_2}, \quad y_3 = \frac{y_1y_2 - ax_1x_2}{1 - dx_1x_2y_1y_2}$$

Neutral element: $O = (0, 1)$.

Inverse: $-(x, y) = (-x, y)$.

Used in: Ed25519 (fast, unified formula).

ECC Domain Parameters

For prime-field curves: $\{p, a, b, G, n, h\}$ where

- p : field prime; a, b : curve coefficients
- G : base point of (large) prime order n
- h : cofactor (small for prime-order subgroups)

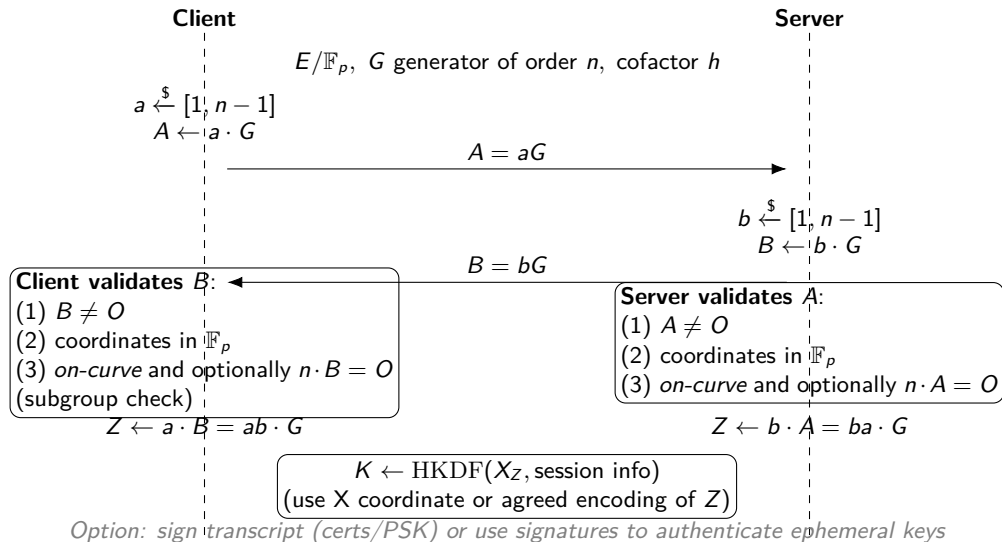
Key pairs: private $d \in [1, n - 1]$, public $Q = dG$.

Validation must-haves:

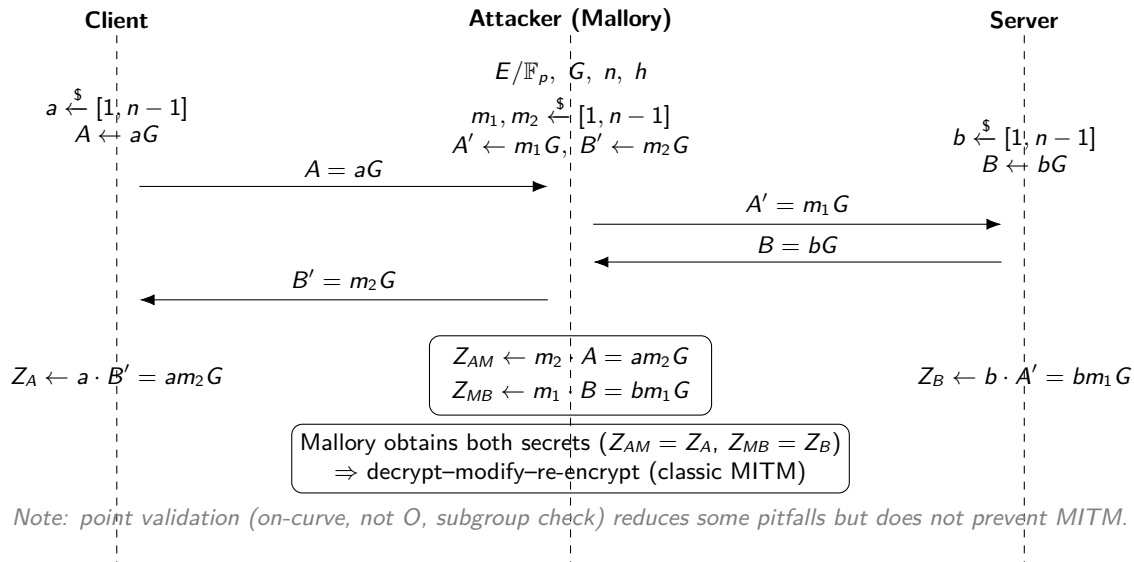
- 1 $Q \neq O$, coordinates in field; satisfies curve equation
- 2 $[n]Q = O$ (on prime-order subgroup)
- 3 Cofactor handling (*clear or clamp*) for Montgomery/Edwards

Thuật ngữ: key pair $(d, Q=dG)$ — cặp khoá $(d, Q=dG)$; public-key validation — kiểm tra khoá công khai; cofactor handling — xử lý hệ số phụ (cofactor)

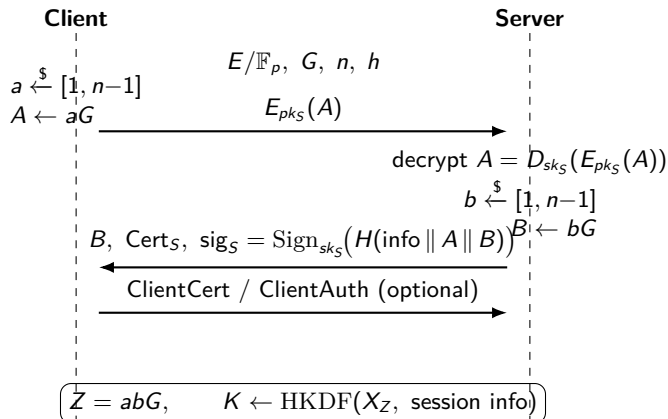
ECDHE Protocol (Ephemeral) — Client / Server



ECDHE — Man-in-the-Middle (No Authentication)

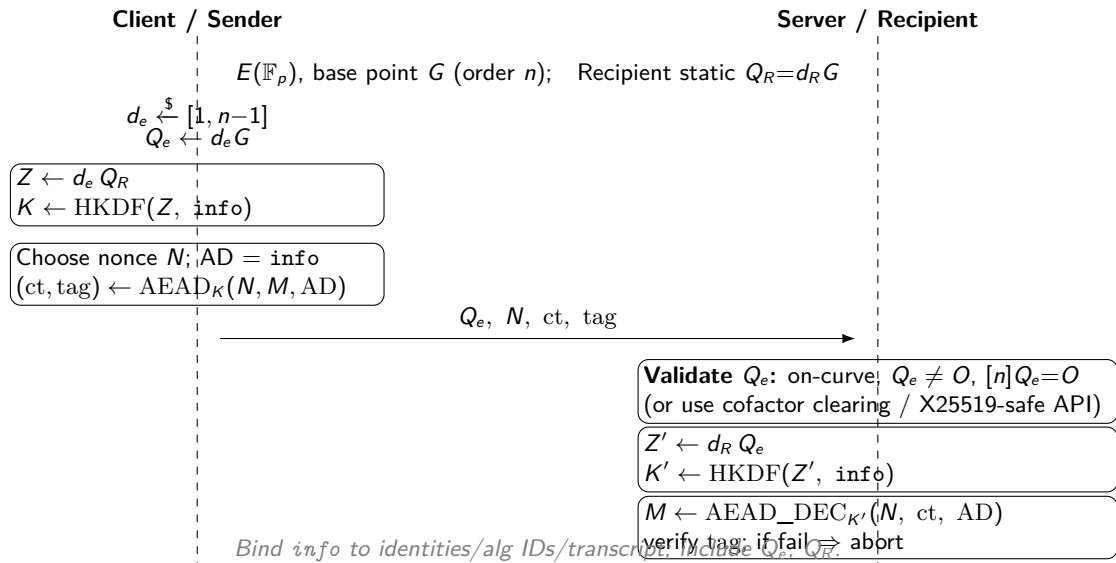


Fix: Authenticated ECDHE — Encrypt Ephemeral Key (Prevent MITM)



Server proves authenticity, Client proves freshness by encrypted A — MITM blocked.

ECIES (KEM-DEM) — Client/Server Figure



Cheat Sheet — Key Sizes & Security Equivalences

AES Security	FFDHE modulus p (bits)	RSA modulus n (bits)	ECC field size (bits) $\frac{1}{2}\log_2(n)$	Typical Use Horizon
112-bit	2048	2048	224 (P-224)	short-term (~ 2030)
128-bit	3072	3072	256 (P-256/X25519)	mid-term ($\sim 2035-2040$)
192-bit	7680	7680	384 (P-384)	long-term (beyond 2040)
256-bit	15360	15360	521 (P-521/Curve448)	very long-term / top-secret

Note: ECC achieves the same bit security with much smaller key sizes (e.g., 256-bit ECC $\approx$ 3072-bit RSA / FFDHE).

Thuật ngữ: Bit security — mức an toàn tính theo số bit tấn công khả thi; FFDHE — Finite-Field Diffie-Hellman; ECC — Elliptic Curve Cryptography; Use horizon — khung thời gian sử dụng khuyến nghị (NIST SP 800-57 Pt 1 Rev 5).

Security Levels & Why ECC

- Similar security with much smaller keys than RSA (e.g., 256-bit ECC $\sim$ 3072-bit RSA).
- Benefits: lower CPU, bandwidth, storage; ideal for constrained devices and high-concurrency servers.
- Choose curves with robust implementations and clear side-channel story (e.g., X25519/Ed25519, P-256).

Thuật ngữ: equivalent security — mức an toàn tương đương; constrained devices — thiết bị hạn chế; safe curves — đường cong an toàn (X25519/Ed25519, P-256)

Implementation Pitfalls (Checklist)

Public-key validation: on-curve, not O , correct subgroup.

Small-subgroup/invalid-curve attacks: do cofactor clearing or validate; prefer complete formulas (Edwards).

Side channels: constant-time field ops, laddering, scalar blinding, cache/timing hardening.

RNG & nonces: deterministic ECDSA or strong CSPRNG; never reuse k .

Protocol binding: KDF includes transcript/context; authenticate ECDH.

Safe APIs: prefer high-level primitives (X25519, Ed25519, ECIES in reputable libs).

Thuật ngữ: public-key validation — kiểm tra khoá công khai; small-subgroup attack — tấn công nhóm con nhỏ; side channels — kênh kề; CSPRNG — bộ sinh số ngẫu nhiên an toàn; protocol binding — ràng buộc giao thức; safe APIs — API an toàn

TLS Integration (Modern Practice)

- TLS 1.2+: ECDHE for key exchange; ECDSA or RSA certificates.
- TLS 1.3: *only* (EC)DHE KEX; separate authentication via certs; session resumption via PSK.
- Cipher suites: AEAD (GCM, ChaCha20-Poly1305); HKDF-based key schedule.

Thuật ngữ: ECDHE — ECDHE; certificates — chứng thư; TLS 1.3 — TLS 1.3; AEAD — mã hoá kèm xác thực; HKDF — HKDF

Post-Quantum Note

- Shor's algorithm breaks integer factoring and ECDLP on large-scale quantum computers.
- Transitional strategy: **hybrid** KEM (X25519 + PQC KEM) and PQ signatures as they standardize/deploy.

Thuật ngữ: Shor's algorithm — thuật toán Shor; hybrid KEM — KEM lai; post-quantum signatures — chữ ký hậu lượng tử

Summary

- ECC delivers strong security with compact keys and efficient operations.
- Use well-vetted curves (X25519/Ed25519, P-256) and constant-time implementations.
- Validate points, handle cofactor, authenticate DH, and use KDF+AEAD.

Thuật ngữ: compact keys — khoá gọn; constant-time — hằng thời gian; point validation — kiểm tra điểm; cofactor handling — xử lý cofactor; KDF+AEAD — KDF+AEAD

Further Reading

- SECG SEC 1/SEC 2: domain parameters and recommended curves.
- NIST SP 800-57 Pt.1: key management and comparable security levels.
- RFC 7748 (X25519/Curve25519) and RFC 8032 (Ed25519).
- SafeCurves: <https://safecurves.cr.yp.to/>

Thuật ngữ: SECG — SECG; NIST — NIST; RFC 7748/8032 — RFC 7748/8032; SafeCurves — SafeCurves

Lattices & Hardness Landscape

Euclidean lattice: $\mathcal{L}(\mathbf{B}) = \{\mathbf{Bz} : \mathbf{z} \in \mathbb{Z}^m\} \subset \mathbb{R}^n$.

Canonical hard problems:

- **SVP/CVP:** find shortest/closest vector; **GapSVP, SIVP**.
- **Worst-to-average reductions:** hardness of certain average-case problems reduces to worst-case (e.g., $\text{LWE} \Rightarrow \text{GapSVP/SIVP}$ up to poly factors).
- **SIS** (Short Integer Solution): find short $\mathbf{x} \neq \mathbf{0}$ s.t. $\mathbf{Ax} \equiv \mathbf{0} \pmod{q}$.

Security intuition: structured noise hides secrets; decoding in high dimension with discrete Gaussians is intractable at chosen parameters.

Thuật ngữ: Trực giác an ninh: nhiễu che giấu bí mật; giải mã trong không gian chiều cao là bất khả thi với tham số phù hợp

From LWE to Ring/Module-LWE

LWE (Learning With Errors): samples $(\mathbf{a}_i, b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \bmod q)$ with small errors e_i ; recover $\mathbf{s}$.

Ring-LWE: lift to polynomial rings $R_q = \mathbb{Z}_q[x]/(x^n + 1)$; secrets/errors in R_q .

Module-LWE: vector (module) over R_q of rank k —trade-off between efficiency and structure; forms the basis of Kyber/ML-KEM.

- Advantages: compact keys, fast NTT-based convolution, IND-CPA with reconciliation, transformed to IND-CCA2 via FO.
- Design knobs: (n, q) , error distribution (e.g., centered binomial), rank k , and rejection/DFR targets.

Thuật ngữ: Từ LWE đến Ring/Module-LWE: mở rộng sang vành đa thức và mô-đun để đạt hiệu năng và gọn nhẹ

Polynomial Arithmetic & NTT (implementation view)

- Choose $n = 256$ and prime $q = 3329$ (Kyber) with suitable primitive roots for radix-2 NTT.
- **NTT**: maps coefficient domain to point-value domain; convolution becomes pointwise multiplication.
- Constant-time sampling (CBD), Barrett/Montgomery reduction, and lazy reduction to minimize branches.
- Side-channel notes: masked sampling, shuffling, constant-time polyvec ops, aligning stack/heap.

Thuật ngữ: Góc nhìn triển khai: NTT giúp nhân đa thức nhanh; chú ý giảm thiểu kênh kề (side-channel)

ML-KEM (Kyber) — Construction Sketch

KEM interface: $\text{KeyGen}(\rightarrow pk, sk), \text{Encaps}(pk \rightarrow ct, K), \text{Decaps}(sk, ct \rightarrow K)$.

- Public matrix via XOF (SHAKE), secrets/errors from centered binomial; compute $\mathbf{u}, \mathbf{v}$; reconciliation yields shared K .
- **FO (Fujisaki–Okamoto) transform:** upgrades IND-CPA PKE to **IND-CCA2** KEM.
- Parameter sets: ML-KEM-512/768/1024 (roughly AES-128/192/256 security targets).

Thuật ngữ: Phác hoạ: KEM với FO để đạt IND-CCA2; ba bộ tham số tương ứng mức an ninh

ML-KEM Parameters Sizes (FIPS 203)

Parameter	Encap Key	Decap Key	Ciphertext	Shared Secret
ML-KEM-512	800	1632	768	32
ML-KEM-768	1184	2400	1088	32
ML-KEM-1024	1568	3168	1568	32

- Decapsulation failure negligible at mandated bounds; FO binding prevents re-encryption malleability.
- Implementation cautions: no RNG reuse across encapsulations; constant-time decapsulation with explicit-failure handling.

Thuật ngữ: Kích thước khoá/bản mã theo FIPS 203; thực thi phải tránh tái sử dụng ngẫu nhiên và cần thời gian hằng

ML-KEM in TLS 1.3

Standalone NamedGroups (pure PQ): MLKEM512, MLKEM768, MLKEM1024 for `key_share` in TLS 1.3.

Hybrid (ECDHE+ML-KEM): drafts define X25519MLKEM768, P-256MLKEM768, P-384MLKEM1024.

- Handshake binds pk and ct via the key schedule; provides re-encapsulation resilience.
- IANA codepoints reserved for ML-KEM NamedGroups; implementations emerging (OpenSSL 3.5+, JDK in progress).
- Operational advice: negotiate hybrids during transition; move to pure ML-KEM when policy permits.

Thuật ngữ: Tích hợp TLS 1.3: nhóm ML-KEM độc lập và các biến thể lai ECDHE+ML-KEM; khuyến nghị vận hành

Threats, Attacks, and Hardening

- **Timing/Cache SCA**: constant-time decapsulation, masked sampling, avoid secret-dependent memory.
- **Faults/DFR abuse**: encode failure-handling to avoid oracle channels; cap retries.
- **KyberSlash/Slash-and-Recover class**: prefer vetted implementations with software/hardware mitigations.
- **Parameter misbinding**: ensure correct ML-KEM level in policy/interop matrices.

Thuật ngữ: Mối đe dọa và gia cố: kênh kề, lỗi gây rò rỉ, triển khai an toàn và điều phối tham số

Mini Lab (for students)

Goal: capture & inspect a TLS 1.3 handshake using ML-KEM (or ECDHE-MLKEM hybrid) and verify `key_share` NamedGroup.

- 1 Build OpenSSL ≥ 3.5 with OQS/ML-KEM support, or use a prebuilt supporting draft ML-KEM groups.
- 2 Start a local TLS server advertising {MLKEM768, hybrid} and connect with a matching client.
- 3 Use tcpdump/Wireshark to verify `key_share` group and ext. sizes; confirm no RNG reuse.
- 4 Report: handshake transcript, group used, sizes vs. FIPS 203 table, and any interop issues.

Thuật ngữ: Bài lab: bắt tay TLS 1.3 với ML-KEM (hoặc lai), xác minh nhóm `key_share` và kích thước

Reading list

- Regev, O. “The Learning with Errors Problem (survey).” (Foundational LWE background).
- FIPS 203: Module-Lattice-Based KEM (ML-KEM) — parameters, sizes, and guidance.
- IETF TLS drafts: ML-KEM NamedGroups; ECDHE-MLKEM hybrids; IANA registry updates.

Thuật ngữ: Tài liệu đọc: tổng quan LWE, chuẩn FIPS 203, bản thảo IETF TLS

Module-LWE and SIS — Formal Definitions

Rings and modules. Fix $n=256$, $q=3329$, $R_q = \mathbb{Z}_q[x]/(x^n+1)$; for rank $k \in \{2, 3, 4\}$ let R_q^k be an R_q -module.

Discrete distributions. Let $D_\eta(R_q)$ denote the distribution of polynomials with i.i.d. coefficients drawn from a centered binomial distribution (CBD) with parameter $\eta \in \{2, 3\}$.

MLWE _{k,n,q,D} (search). Given i.i.d. samples $(\mathbf{a}_i, b_i) \in R_q^k \times R_q$ with

$$b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \in R_q, \quad \mathbf{s} \leftarrow D_\eta(R_q)^k, \quad e_i \leftarrow D_\eta(R_q),$$

recover $\mathbf{s}$.

MLWE (decision). Distinguish $(\mathbf{a}_i, \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i)$ from uniform $(\mathbf{a}_i, u_i) \in R_q^k \times R_q$.

SIS _{n,q,β} (short integer solution). Given $A \in \mathbb{Z}_q^{m \times n}$, find nonzero $\mathbf{x} \in \mathbb{Z}^n$ with $\|\mathbf{x}\| \leq \beta$ s.t. $A\mathbf{x} \equiv \mathbf{0} \pmod{q}$.

Hardness (informal). For suitable (n, q, k, η) , average-case MLWE reduces from worst-case ideal/module lattice problems (e.g., GapSVP/SIVP), which underpins ML-KEM's security parameters.

Thuật ngữ: Định nghĩa hình thức và cơ sở độ khó

Encodings: Compression, Decompression, and CBD Sampling

Coefficient compression ($x \in \mathbb{Z}_q$):

$$\text{Compress}_d(x) = \left\lfloor \frac{2^d}{q} \cdot (x \bmod q) \right\rfloor \in \{0, \dots, 2^d - 1\}, \quad \text{Decompress}_d(y) = \left\lfloor \frac{q}{2^d} \cdot y \right\rfloor \in \mathbb{Z}_q.$$

Apply coefficient-wise to polynomials/vectors.

Centered Binomial Distribution (CBD_η). From bytes B , split into 2η sub-chunks; for each coefficient

$$e = \sum_{i=1}^{\eta} \text{Hwt}(b_i) - \sum_{i=\eta+1}^{2\eta} \text{Hwt}(b_i) \in \{-\eta, \dots, \eta\},$$

yielding subgaussian noise with variance $\approx \eta/2$ (centered).

Thuật ngữ: Nén/giải nén hệ số; mẫu nhiễu CBD

Negacyclic Convolution via NTT

Let $\zeta \in \mathbb{Z}_q$ be a primitive $2n$ -th root with $\zeta^{2n} \equiv 1 \pmod{q}$ and $\zeta^n \equiv -1$. For $f \in R_q$, define

$$\widehat{f}[i] = f(\zeta^{2i+1}) = \sum_{j=0}^{n-1} f_j (\zeta^{2i+1})^j \pmod{q}, \quad i = 0, \dots, n-1.$$

For $h = f * g \pmod{x^n+1}$, pointwise multiplication gives

$$\widehat{h}[i] = \widehat{f}[i]\widehat{g}[i] \pmod{q}, \quad h = \text{NTT}^{-1}(\widehat{h}).$$

Implementation: twiddle factors with bit-reversed indexing; Montgomery/Barrett reductions; constant-time butterflies.

Thuật ngữ: Tích chập negacyclic qua NTT

K-PKE (CPA PKE over Module-LWE) — Algebraic Form

KeyGen. Sample ρ, σ via XOF; derive $\mathbf{A} \in R_q^{k \times k}$ from ρ . Sample $\mathbf{s} \leftarrow D_{\eta_1}(R_q)^k$, $\mathbf{e} \leftarrow D_{\eta_1}(R_q)^k$. Compute

$$\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e} \in R_q^k.$$

Publish (compressed) $(\rho, \mathbf{t})$; keep $\mathbf{s}$ (and required tags).

Encrypt. For $m \in \{0, 1\}^{256}$ (mapped to R_q), sample $\mathbf{r} \leftarrow D_{\eta_1}(R_q)^k$, $\mathbf{e}_1 \leftarrow D_{\eta_2}(R_q)^k$, $e_2 \leftarrow D_{\eta_2}(R_q)$. Compute

$$\mathbf{u} = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 \in R_q^k, \quad v = \langle \mathbf{t}, \mathbf{r} \rangle + e_2 + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor \in R_q,$$

where μ encodes the bits of m . Output $c = (\text{Compress}_{d_u}(\mathbf{u}), \text{Compress}_{d_v}(v))$.

Decrypt. Compute $v' = v - \langle \mathbf{u}, \mathbf{s} \rangle \in R_q$ using decompressed values; recover μ by thresholding near $\lfloor q/2 \rfloor$; invert the message encoding.

Thuật ngữ: Dạng đại số PKE: $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$; mã hoá/giải mã với nén hệ số

From K-PKE to ML-KEM (FO Transform, IND-CCA2)

Let H, G be SHAKE-based XOF/PRF. Denote CPA PKE above by PKE.

KeyGen : $(ek, dk) \leftarrow \text{PKE.KeyGen}$; cache $H(ek)$.

Encaps : $m \xleftarrow{\$} \{0, 1\}^{256}$; $(c, \tilde{K}) \leftarrow \text{PKE.Encrypt}(ek, m; r=G(m\|H(ek)))$;
 $K = H(m\|c)$.

Decaps : $m' \leftarrow \text{PKE.Decrypt}(dk, c)$; $r' = G(m'\|H(ek))$; $(c', \tilde{K}') = \text{PKE.Encrypt}(ek, m'; r')$;
if $c' = c$ **then** $K = H(m'\|c)$ **else** $K = H(z\|c)$,

where z is a secret fallback in dk . All checks are constant-time. This yields IND-CCA2 in the RO model under MLWE.

Thuật ngữ: Biến đổi FO: kiểm thử tái mã hoá chống CCA

Correctness and Decapsulation Failure

With well-formed inputs and correct bounds, noise margins from (d_u, d_v) ensure negligible failure:

$$\Pr[\text{Decaps fails}] \leq 2^{-138.8} (512), \quad 2^{-164.8} (768), \quad 2^{-174.8} (1024).$$

Bounds track coefficient growth across NTT multiplies and decode regions.

Thuật ngữ: Xác suất thất bại là siêu nhỏ

Implementation Notes: Constant-Time and SCA

- All comparisons ($c' \stackrel{?}{=} c$), conditional moves, and rejects are constant-time.
- Domain-separated XOF for **A**; never reuse r across encapsulations.
- NTT/INTT butterflies with Montgomery reduction; avoid secret-dependent memory.
- Bound rejection sampling loops (SampleNTT) to analyzed limits for negligible DFR.

Thuật ngữ: Ghi chú triển khai: thời gian hằng; tránh kênh kề

TLS 1.3 Mapping (Quick Math Pointers)

ClientHello: `key_share` carries ek ; ServerHello returns c . Transcript hash binds (ek, c) ; KEM shared secret K feeds TLS HKDF schedule. Validate $|ek|, |c|$ against ML-KEM sizes table.

Thuật ngữ: Ảnh xạ TLS: ràng buộc bản ghi; kiểm kích thước

Thank you!